Policy Brief: Where District Development Lags in India (NFHS-5)

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1. Problem

Public spending works best when it reaches the districts that need it most, and in the sector where they are weakest. District health and education data exist, but they sit in separate tables and are hard to compare. This brief combines them into one District Development Index to show which districts are furthest behind, and whether their main gap is in health or in education.

2. Data and method

- **Source:** NFHS-5 (2019-21) district fact sheets, as converted to CSV in a community GitHub repository. These are unofficial conversions, and a few districts were spot-checked against the official sheets.
- **Coverage:** 642 districts across 37 states and union territories.
- **Indicators (7):** stunting, underweight, institutional births, full immunisation, female literacy, female population that ever attended school, and women with 10 or more years of schooling.
- **Index:** each indicator was scaled from 0 to 1, with stunting and underweight reversed so that higher always means better. Indicators were averaged into a health score and an education score, and the two were averaged into a 0-100 index with equal weights.
- **Robustness check:** the ranking was rebuilt with weights from principal component analysis. The two rankings are almost identical (Spearman correlation 0.993), and 48 of the 50 lowest-ranked districts are the same under both methods. The average district moved about 16 places out of 642.
- **Grouping:** K-Means clustering on the health and education scores gave three groups, chosen using the silhouette score.

3. Findings

Three groups of districts.

Group	Districts	Avg health	Avg education	Avg index
Higher-performing	197	0.78	0.72	75
Middle	278	0.68	0.52	60
Priority: weak on both	167	0.53	0.36	44

About 26% of scored districts fall in the Priority group.

Priority districts are concentrated in a few states. Bihar has 33 of its 38 districts in the Priority group (87%), and Jharkhand has 16 of 23 (70%). Uttar Pradesh (33 of 71) and Madhya Pradesh (23 of 50) each have 46%. These four states hold 105 of the 167 Priority districts, about 63%.

The lowest-ranked districts. The bottom ten are Bahraich, Pakur, Araria, Shrawasti, Mewat, Pashchimi Singhbhum, Bijapur, Sahibganj, Purnia and Balrampur. The top ten are mostly in Kerala and Tamil Nadu, led by Pathanamthitta, Kottayam and Kanniyakumari.

Districts have different kinds of problems.

- *Education trails health:* Mewat (Haryana) scores 0.43 on health but 0.17 on education. Shrawasti (health 0.42, education 0.15) and Bahraich (0.36 and 0.14) show a similar pattern.
- *Health trails education:* in parts of the North-East, such as Zunheboto (health 0.30, education 0.67), Tuensang (0.30 and 0.54) and West Khasi Hills (0.31 and 0.56), education is much stronger than health.

4. Recommendations

1. **Treat Bihar and Jharkhand as an integrated-package priority.** Most of their districts are weak on both health and education, so a single-sector programme is unlikely to be enough.
2. **Start with the lowest-scoring districts in Uttar Pradesh and Madhya Pradesh.** They have the largest number of Priority districts in absolute terms. Bahraich, Balrampur and Shrawasti are the first places to look.
3. **Put literacy and schooling first where education lags health.** In districts like Mewat, Shrawasti and Bahraich, more clinics are not the main gap. Teacher availability, school retention and female literacy programmes are.
4. **Put health services first where health lags education.** In the North-East districts above, institutional births, immunisation and child nutrition services deserve attention. Verify with local data first, because district estimates there rest on small samples.

5. **Use the index to screen, not to decide.** Combine it with school dropout data (UDISE+) and employment data before allocating funds.

5. Limitations

- The index covers health and education only. It has no income, employment or infrastructure indicators.
- NFHS-5 data are from 2019-21 and may not reflect the present situation.
- District-level survey estimates have sampling error, especially in small districts. Small states can show large percentage swings, for example Nagaland, where 5 of 11 districts are in the Priority group.
- Health and education scores move together, so the three groups are mostly levels of development, not distinct types of problem.
- The district data are community conversions of the official PDFs, with limited checking.
- Equal weights are a judgement. The sensitivity check shows the ranking is stable, but it does not show that equal weights are the right choice.

6. Supporting material

An interactive three-page dashboard (overview, state comparison and district rankings), the notebooks and the data sources are available in the project's GitHub repository.